BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, DI KHAN

INTERMEDIATE (12TH CLASS) EXAMINATION, 2021

BIOLOGY

Past Paper 2021 • Paper Code: DIK-HSSC-XII-2021 (A)

Retyped faithfully from the original board paper — same questions, no changes.

SECTION – A

Time Allowed: 20 min | **Maximum Marks:** 20 | **Paper Code:** DIK-HSSC-XII-2021 (A)

Q1. Choose the correct option i.e. A,B,C,D and fill it up in the correct bubble in the given MCQs sheet.

Qi. The adrenal cortex produces _____

(A) Adrenaline
(B) Calcitonin
(C) Epirephrim
(D) Aldosterone

Qii. The random loss of allele in a population is called _____

(A) Mutatoni
(B) Selection
(C) Gene flow
(D) Genetic drift

Qiii. The enzyme DNA polymerase can work only in _____

(A) 3′−5′
(B) 5′−3′
(C) Both of direction
(D) None of these

Qiv. The red coloured of the muscles fibres is due to _____

(A) Haemolgobin
(B) Myoglobin
(C) Iron
(D) Calcium

Qv. Greenhouse effect is the result of _____

(A) Industrialization
(B) Deforestation
(C) Over urbanization
(D) All of these

Qvi. The behavior by which animals are arranged according to their status is called _____

(A) Altruism
(B) Dominance Hierarchies
(C) Agonistic behavior
(D) Territorial behavior

Qvii. In Bombay phenotype blood group is phenotypically like___

(A) AB
(B) A
(C) B
(D) O

Qviii. Oxytocin hormones is produced by _____ gland.

(A) Thyroid
(B) Parathyroid
(C) Adrenal
(D) Pituitary

Qix. Effector may be _____

(A) Gland
(B) Bone
(C) Muscle
(D) Both A and C

Qx. Pneumonia is caused by _____

(A) Bacteria
(B) Virus
(C) Mycoplasma
(D) All of these

Qxi. The number of linkage group in man is _____

(A) 2
(B) 32
(C) 46
(D) 92

Qxii. The corpus luteum is formed at the site of _____

(A) Fertilization
(B) Ovulation
(C) Menstruation
(D) Implantation

Qxiii. Which of the following is a denitrifying Bacteria?

(A) Rhizobium
(B) Pseudomonas
(C) Mycoplasma
(D) All of these

Qxiv. Which one is the most essential method of water loss in the body?

(A) Excretion in urine
(B) Sneezing
(C) Sweating
(D) Breathing

Qxv. Which hormone is amino acid derivative?

(A) Thyroxin
(B) ADH
(C) Progesterone
(D) Estrogen

Qxvi. Which of the following papillae do not contain taste buds?

(A) Filiform
(B) Fungiform
(C) Circumvallate
(D) Foliate

Qxvii. Human lungs are spongy due to presence of millions of___

(A) Trachea
(B) Bronchioles
(C) Alveoli
(D) Bronchi

Qxviii. _____ is a start codon.

(A) AUG
(B) UGA
(C) UAG
(D) UAA

SECTION – B

Time Allowed: 2:40 Hrs | **Maximum Marks:** 40

Q2. Attempt any Ten questions of the following. All questions carry equal marks

i. What are the symptoms and treatment of pneumonia?

ii. What are the salient feature of Darwin-Wallace theory?

iii. Differentiate between cortical nephron and juxtramedullary nephron.

iv. Define deletion and insertion.

v. Differentiate between primary succession and secondary succession.

vi. Define and discuss sinusitis.

vii. What is greenhouse effect? Write its consequences.

viii. Differentiate miscarriage and abortion.

ix. Draw a label diagram of a typical neuron.

x. Define and exemplify Altruistic behavior.

xi. Give difference between haemoglobin and myoglobin.

xii. Differentiate between dominance and epistasis.

xiii. Write down the significant rule of insulin.

SECTION – C

Maximum Marks: 27

Note: Note: Attempt any three Question. Each question carries equal marks.

Q.3 Describe polymerase chain reaction (PCR) in detail. Write down its application.

Q.4 Write note on any two of the following: (a) Sex-determination (b) What are the properties of Genetic code? (c) Ecological succession

Q.5 Define xerosores. Discuss various stages of xerosares on lithosares.

Q.6 How Mendel performed dihybrid cross?